

Answer Key — Biology Grade 10

Sub-Strand 2.1: Plant Nutrition • Lesson 2: Leaf Structure: The Architecture of a Photosynthetic Machine

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5
A	B	B	B	B

USE AFTER: OBSERVE PHASE — Step 1: whole leaf under the hand lens (upper vs lower surface)

Q1. Why is it useful that the cuticle is both waxy and transparent?

- A. It lets light in while reducing water loss**
- B. It traps carbon dioxide inside the leaf
- C. It makes food for the leaf
- D. It carries water to the leaf cells

Correct answer: A. Transparent lets light reach the chloroplast-rich cells below; waxy slows water loss. The cuticle does not photosynthesise or transport water.

USE AFTER: OBSERVE PHASE — Step 2: cross-section under the microscope

Q2. Which layer is packed tightly with chloroplasts to capture the most light?

- A. Lower epidermis
- B. Palisade mesophyll**
- C. Spongy mesophyll
- D. Cuticle

Correct answer: B. Palisade cells are tall and tightly packed under the upper epidermis, where light is strongest. Spongy mesophyll is loosely packed for gas movement.

USE AFTER: OBSERVE PHASE — Step 4: locating stomata on the lower epidermis

Q3. What controls how wide a stoma opens?

- A. The cuticle
- B. The two guard cells around it**
- C. The palisade cells
- D. The xylem vessels

Correct answer: B. Each stoma is surrounded by two guard cells that change shape to open or close the pore. HOW they do this is left as a DQB question for later in the unit.

USE AFTER: OBSERVE PHASE — Step 3: labelling the layers on the outline diagram (vascular bundle)

Q4. Which tissue carries water from the roots up into the leaf?

- A. Phloem
- B. Xylem**
- C. Palisade mesophyll
- D. Epidermis

Correct answer: B. Xylem carries water and minerals up from the roots. Phloem carries sugars made in the leaf to the rest of the plant.

Q5. Using today's evidence, why does the sukuma wiki wilt at midday even though the soil is moist?

- A. The soil dries out faster in strong sunlight
- B. Water vapour leaves through open stomata faster than xylem can replace it**
- C. The roots stop absorbing water when it is hot
- D. Chlorophyll breaks down in direct sunlight

Correct answer: B. The lesson's payoff: open stomata in bright light lose water vapour faster than the xylem resupplies it, so cells lose water and the leaf droops. The soil stays moist, so A and C do not fit the phenomenon.